

ARYAN HINGWE

+91 9325546348 | arianhingwe1612@gmail.com | github.com/aryanhingwe

Course	Examination	University	Major	Year	CGPA /10
Integrated BS-MS	Post Graduation	IISc Bangalore	Physics	2024-25	8.0
	Graduation	IISc Bangalore	Physics	2020-24	8.1

AWARDS AND ACHIEVEMENTS

- **JEE Mains and JEE Advanced exams:** Secured an **All India Rank 388** in IIT JEE Main 2020 amongst more than **1 million** students and **All India Rank 1313** in IIT JEE Advanced 2020 amongst more than **0.2 million** students
- **KVPY Fellow:** Achieved an **All India Rank 60** in KVPY exam and awarded KVPY scholarship by Department of Science and Technology, Government of India. Secured admission at the Indian Institute of Science from the merit list
- **WorldQuant International Quant Championship 2026:** Qualified Level 1 in the **top 1%** of teams internationally. Among the selected teams in level 2, achieved **top 2%** in the final overall international ranking.

EXPERIENCE | RESEARCH WORK

- **Modelling risk-propagation in ETFs using modified multivariate Hawkes process**
| *Guides: Prof. Shashi Jain & Prof. Srikanth Iyer* (June '25 – May '26)
 - This project develops a modified multivariate Hawkes process to model and analyse ETF-driven risk propagation among underlying assets at high frequencies
 - This work is under review
- **WorldQuant BRAIN Research Consultant** (June '26 – Ongoing)
 - Research Consultant: 35+ signals with high sharpe and fitness generated and submitted

PROJECTS

- **Option Pricing using the Heston, Black-Scholes-Merton, and Binomial Tree Models** (Jan '25 – Apr '25)
 - Calibrated parameters to the historical NSE stock and options prices, implemented a hedging strategy over multiple periods, and analysed price movements using the Greeks- Delta, Theta, Gamma, and Vega
- **Portfolio Optimization and Factor Analysis – Markowitz & Fama-French Models** (Aug '24 – Dec '24)
 - Developed a portfolio optimization pipeline for daily returns to compute the Markowitz efficient frontier, optimal weights and risk measures- Sharpe ratio, VaR, CVaR and Maximum Drawdown for equity portfolios
 - Optimized allocations across 200+ equities; backtesting showed Sharpe ratio maximizing portfolio performing the best in achieving returns of **22%YoY (Year on Year)** and β (beta) of **1.32**
 - Performed **Fama-French regression** to analyze exposure to momentum, size (SMB), and value (HML) risk factors.
- **Natural Language Processing tool for genre classification of news headlines** (Jan '24 – April '24)
 - Implemented TF-IDF analysis for tokenization of text and used a multilayer neural network to get an accuracy of **83%** for classification of **210,294** news headlines with genre- travel, business, etc from HuffPost
 - Utilised gradient descent, regularisation and hyperparameter tuning to maximize predictive accuracy

SELECTED COURSES & CERTIFICATIONS

- **Bloomberg:** Market Concepts, Finance Fundamentals, and Spreadsheet/BQL certificates
- **Programming courses:** Pattern Recognition and Neural Networks, Algorithms And Programming, Advanced Learning Algorithms in Deep learning on Coursera
- **Finance courses:** Corporate Finance, Financial instruments and Risk management strategies, Micro Economics
- **Mathematics and Statistics:** Linear Algebra and Complex Analysis, Probability and Statistics, Mathematical Methods in Physics- Ordinary and Partial differential equation, 3d Vector analysis, Complex variables

TECHNICAL SKILLS

- **Programming Languages & Tools:** Python, C++, Linux, Excel, Bloomberg Query Language, $\text{\LaTeX}$, Mathematica
- **Python libraries:** Numpy, Pandas, Scipy, Statsmodels, Scikit-learn, PyTorch, Matplotlib, Seaborn
- **Numerical methods:** Time Series, Machine Learning, Neural Networks, Monte Carlo methods, Differential equations

POSITIONS OF RESPONSIBILITIES & EXTRA-CURRICULAR ACTIVITIES

- **Recent Trends in Quantitative Finance 2026 at IISc:** Part of the organising committee for the two day symposium
- **Sponsorship Core Coordinator | Rhapsody 23**
 - Responsible for heading fundraising for the annual IISc student mega-fest with a footfall of **10,000+** people
 - **Strategising, negotiations and smooth conduction of deliverables** for **30+** sponsors across all events of Rhapsody
- **Teaching Assistant** for the courses: 1. Financial instruments and Risk management strategies 2. Corporate finance, evaluated exams and conducted office hours to assist students.